**Solar ESS Co. Ltd.**
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Tel 063 151 6694 · 089 632 5829
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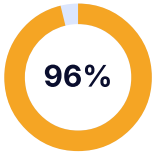
PLANT PERFORMANCE REPORT

# Chicken farm, Chiang Rai

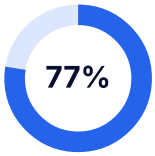
20 kWp grid-tied rooftop system · 20 kW inverter · no battery · Chiang Rai

| PERIOD | DATA SOURCE | ISSUED |
|---|---|---|
| 1 – 30 October 2026 (30 days) | Plant monitoring platform | 31 October 2026 |

| Solar production | Average per day | Farm consumption | Bought from grid | Sent to grid | Specific yield |
|---|---|---|---|---|---|
| 1,458.0 kWh | 48.6 kWh/day | 1,807 kWh | 407.8 kWh | 58.5 kWh | 2.43 kWh/kWp/day |




of solar used on the farm

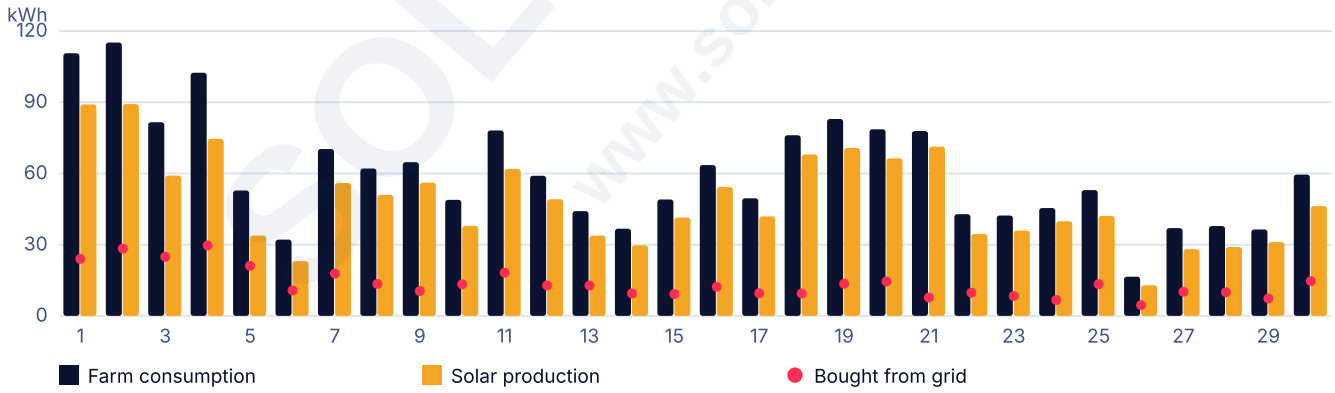



of the farm's electricity came from solar

## Summary

- The system produced **1,458.0 kWh** in October and the farm used **1,807 kWh**. Solar covered **77%** of the farm's electricity.
- Almost all solar was used on site (**96%**). Only about **2 kWh a day** went to the grid.
- Average output was **2.43 kWh per kWp per day**, well below what a 20 kWp array can give. The main reason: the inverter is **holding production down to match the farm's load** (export limit).

## Daily production, consumption and purchases




Production rises and falls with the farm's consumption rather than with the weather: on 1–2 October the farm used about 110 kWh and the panels made 89 kWh; on light-load days the panels made only 13–35 kWh.

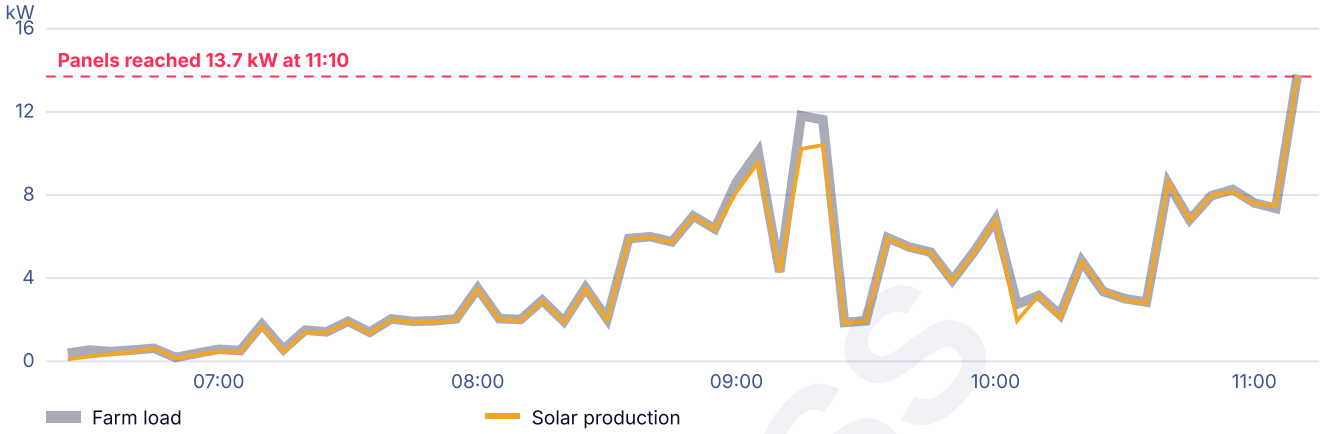
## Findings

1. **Output is capped at the farm's load.** The export limit keeps production equal to consumption. On sunny days an estimated 20–30 kWh a day of solar is not harvested.
2. **Most grid purchases are at night.** The farm buys about 14 kWh a day, mainly overnight (base load 0.2–0.5 kW with regular 0.7–1.1 kW pump or fan cycles) and in the early morning.
3. **Monitoring gaps.** Data for January, February and August is incomplete. The logger connection should be checked so no month is lost again.
4. **No alarms.** The inverter is online with no active alerts.

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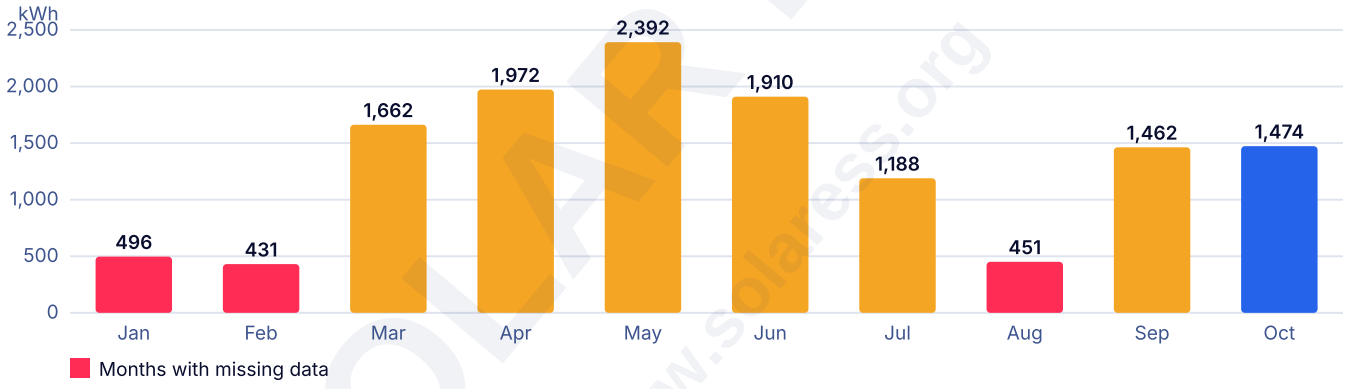
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## This morning: production follows the load




31 October, 06:25–11:10, 5-minute data. The solar line sits on top of the load line all morning, within about 40 W. When the load jumped at 11:10, production rose at once to 13.7 kW: the panels can give much more than they are allowed to.

## Production by month, 2026




January, February and August are far too low for this system. The inverter's own counter shows about 23,085 kWh lifetime, while the platform shows 14.61 MWh: the data logger was most likely offline in those months.

### Recommendations

1. **Review the export limit.** If the farm has a sell-back agreement, raise the limit so surplus solar earns income. If not, keep the limit.
2. **Use more power in daytime.** Run pumps, fans, water heating and feed equipment between 09:00 and 15:00 where possible.
3. **Consider a battery of about 10–15 kWh.** It would store the midday solar that is now held back and cover most of the night purchases (if the inverter is a hybrid model, to be confirmed on site).
4. **Check the data logger** Wi-Fi signal and power supply.
5. **Panel cleaning** cannot be judged while output is capped: check again after the limit is changed.

Notes: figures for 1–30 October (full days). Solar used on site = production − sent to grid. Curtailment estimate based on the 13.7 kW reached at 11:10 and the farm's best months (May: 3.9 kWh/kWp/day).